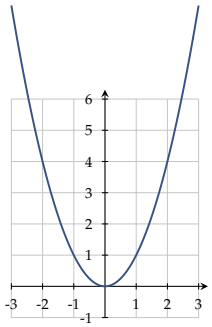
Sheet 22 Thu 10 Dec. 2026 *Square root and cube; sign of reference functions*

1. Warm-up · Quick Calculate $\sqrt{49}$, $\sqrt{0.25}$ and $(-2)^3$.
2. Core · Method Study the sign of $f(x) = -2x + 6$.
3. Core + · True/false True or false? Justify. For every real number x, $\sqrt{x^2} = x$.
4. Review · Quick How many squares make up a net of a cube? Does a cross (a row of 4 squares + 1 square on each side) work?
5. Thread · Quick How many faces, vertices and edges does a square-based pyramid have?

Sheet 23 Mon 14 Dec. 2026 *Graphical and numerical solving: $f(x) = k$, $f(x) < k$*

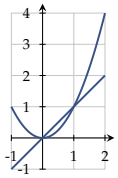
1. Warm-up · Quick Solve $x^2 = 16$ then $x^2 = -4$.
2. Core · Graph Curve of $f(x) = x^2$:



   Solve graphically $f(x) = 4$ then $f(x) < 4$.
3. Core + · Method Solve $|x| < 3$ then $|x| = 5$.
4. Review · Pitfall Calculate $-x^2$ then $(-x)^2$ for $x = -3$.
5. Thread · Method In a triangle ABC, M and N are the midpoints of $[AB]$ and $[AC]$. What is being proved? Which property is cited?

Sheet 24 Tue 15 Dec. 2026 *Solving $f(x) = g(x)$ and $f(x) < g(x)$*

1. Warm-up · Quick Solve $2x - 1 = x + 3$.
2. Core · Graph Curves of $f(x) = x$ and $g(x) = x^2$:



   Solve $f(x) = g(x)$ then $f(x) < g(x)$.
3. Core + · Code reading What does `f(-1) == g(-1)` return? And `f(0) == g(0)`?

```
def f(x):
    return x*x
def g(x):
    return 2*x + 3
```

4. Review · Problem Plan A: 15 000 Ar fixed + 1 000 Ar per session. Plan B: 2 500 Ar per session. From how many sessions is A cheaper?
5. Thread · Quick Calculate the coordinates of the midpoint of $[AB]$ with $A(-2;5)$ and $B(6;1)$.

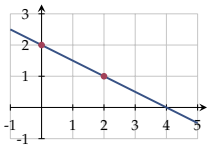
One-variable descriptive statistics

Sheet 25 Thu 17 Dec. 2026 *Mean and linearity of the mean*

1. Warm-up · Quick Mean of 12, 15, 9, 14?
2. Core · Method The series 4, 6, 8, 10, 12 has mean 8. What is the mean if 3 is added to each value? If each value is multiplied by 2?
3. Core + · True/false True or false? Justify. The mean of two numbers $a < b$ is always strictly between a and b.
4. Review · Reading In a pie chart of 80 people, one sector represents 25 %. How many people does it represent? What is its angle?
5. Thread · Quick Calculate PGCD(84 ; 126) and deduce the simplest form of $\frac{84}{126}$.

Sheet 26 Mon 4 Jan. 2027 *Median and quartiles*

1. Warm-up · Quick Order 7, 2, 9, 4, 5, 11, 3 then give the median.
2. Core · Method Series 3 ; 5 ; 6 ; 8 ; 9 ; 12 ; 14 ; 15. Give Q_1, the median, Q_3 and the interquartile range.
3. Core + · True/false True or false? Justify. At least half of the values of a series are greater than or equal to its median.
4. Review · Quick Put 0.6, $\frac{5}{8}$ and 61 % in increasing order.
5. Thread · Graph → formula The line below represents an affine function f. Determine $f(x)$.



Sheet 27 Tue 5 Jan. 2027 *Standard deviation; effect of adding or removing a value*

1. Warm-up · Quick Calculate $(3-5)^2 + (7-5)^2$.
2. Core · Method Series 2 ; 4 ; 6: mean, variance and standard deviation (approximate value to 10^{-2}).
3. Core + · True/false True or false? Justify. Adding 5 to each value of a series does not change its standard deviation.
4. Review · Construction Let ABC be a triangle, I the midpoint of $[AB]$ and D the reflection of C in I. What type of quadrilateral is $ACBD$?
5. Thread · Quick $(MN) \parallel (BC)$, $M \in [AB]$, $N \in [AC]$, $AM = 3$, $AB = 5$, $AC = 10$. Calculate AN.

Sheet 28 Thu 7 Jan. 2027 *Comparing series*

1. Warm-up · Quick Range of 3, 8, 5, 12?
2. Core · Interpret Two series A and B have the same mean 12; $\sigma_A = 1.5$ and $\sigma_B = 4$. Which one is more consistent?
3. Core + · Interpret A: median 11, interquartile range 2. B: median 13, interquartile range 6. Compare the two series.
4. Review · Quick Two fair coins are tossed. What is the probability of getting two heads?
5. Thread · Method Programme: choose a number; square it; add 1. Result for -3? Can 0 be obtained? Justify.